THARUN KUMARAN M

+91-9941876113 | tharunkumaranm@outlook.com | linkedin.com/in/tharunkumaran11 | github.com/tharunkumaran11

Professional Summary

Aspiring Cloud & DevOps Engineer who automates the path from code to production on AWS. Combines networking foundation with hands-on work in Kubernetes, CI/CD, and Infrastructure as Code, and is looking for a DevOps role where I can build and operate scalable, well-monitored cloud environments.

Career Highlights

- Hands-on experience building and deploying **AWS cloud and DevOps projects** covering CI/CD, GitOps, Kubernetes, monitoring, networking, and cloud infrastructure.
- Practical experience with **Amazon EKS, Docker, Helm, Argo CD, GitHub Actions, Jenkins, Prometheus, Grafana, and AWS services**, developed through self-built and fully deployed projects.
- Strong foundation in **Linux, networking, TCP/IP, DNS, VPC, IAM, and load balancing**, supported by an Electronics & Communication Engineering background.
- Demonstrated ability to **design, automate, troubleshoot, and deploy complete cloud environments** independently, with projects documented and maintained on GitHub.

Technical Skills

- **Cloud & Infrastructure:** AWS, Terraform (IaC), EC2, RDS, Lambda, VPC, IAM, S3, EKS, Fargate, Application Load Balancer (ALB), Auto Scaling, CloudWatch, AWS CLI, SNS, SQS
- **Containers & Orchestration:** Docker, Kubernetes, Amazon EKS, AWS Fargate, Helm, Docker Compose
- **CI/CD & Configuration Management:** Git, GitHub, GitHub Actions, Jenkins, Ansible, Argo CD, GitOps, CI/CD Pipeline Design
- **Monitoring & Observability:** Prometheus, Grafana, Node Exporter, Amazon CloudWatch
- **Networking:** TCP/IP, DNS, Subnetting, VPC Networking, Security Groups, Load Balancing, Nginx
- **Linux & Scripting:** Linux, Bash, Python, YAML, JSON

Project Experience

AWS Secure DevOps Platform: VPN-Secured Multi-Server Architecture on AWS GitHub

- Architected a 3-server AWS environment with separate VPN/authentication, application upload, and Jenkins/monitoring servers, securing internal application access through OpenVPN.
- Automated file routing with Jenkins, sending 5 document/image formats to Amazon S3 and 4 code/configuration formats to GitHub based on file type.
- Implemented infrastructure monitoring using Prometheus, Node Exporter, and Grafana, tracking CPU, RAM, disk, and system load across the environment.
- Configured 3 CloudWatch alarms for CPU utilization, Jenkins health, and SQS backlog with SNS email notifications for operational alerts.

Go Web Application – End-to-End CI/CD & GitOps Pipeline on AWS EKS GitHub

- Built and tested a Go web application, containerized using a multi-stage Docker build and tagged images with Git commit SHA for version traceability.
- Automated CI with GitHub Actions to run tests, build and push Docker images to Docker Hub, and update the Helm image tag on every push to the main branch.
- Implemented GitOps-based continuous delivery with Argo CD, automatically synchronizing Helm deployments from GitHub to Amazon EKS without manual kubectl apply.
- Exposed the application through Kubernetes Ingress and the AWS Load Balancer Controller, provisioning an internet-facing Application Load Balancer for external access.

2048 Game on Amazon EKS – Serverless Kubernetes with Fargate & ALB GitHub

- Deployed the 2048 application on Amazon EKS with AWS Fargate, using serverless Pod compute without managing EC2 worker nodes.
- Configured the application with 5 Kubernetes replicas and verified all 5/5 Pods healthy across an EKS environment spanning 3 Availability Zones.
- Built the networking environment with 6 subnets (3 public and 3 private) and configured Internet Gateway and NAT Gateway connectivity.
- Configured the AWS Load Balancer Controller with IRSA to provision an internet-facing Application Load Balancer through Kubernetes Ingress.

Education & Certifications

B.E. Electronics and Communication Engineering –
Dhanalakshmi College of Engineering, Chennai 2021 – 2025
CGPA: 7.66 | HSC: 81.99% | SSLC: 77.8%
Certifications: AWS Training, Linux Training, DevOps Training, CCNA Training